

Predicting the Zero-Shot Transfer of a Promptable Video Tracker

Forecasting SAM 3's reliability on unseen species & places — from label-free
distances

Kostiantyn Boiar

MSc Thesis Proposal ▪ University of Glasgow

Supervisor: Dr Marwa Mahmoud

July 7, 2026

The problem

Promptable trackers (**SAM 3**) segment & track *any* named animal in a video with **no training on that species**.

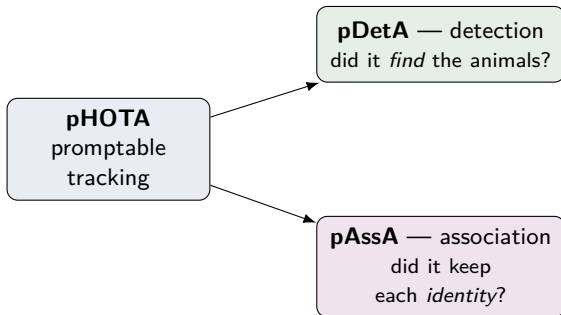
- Conservation teams point them at new species / regions **constantly**.
- But nobody can say *in advance* whether the model is **trustworthy** for a given animal in a given place.
- There is **no rule** that predicts this. That gap is what we fill.

The question

Can we predict SAM 3's zero-shot accuracy on an **unseen (species, place)** *before running it* — from properties measurable in advance?

- **Not** trying to make the tracker score higher — we predict & explain its reliability.
- And: *which factors* govern transfer, separately for **finding** the animal and **following** it?

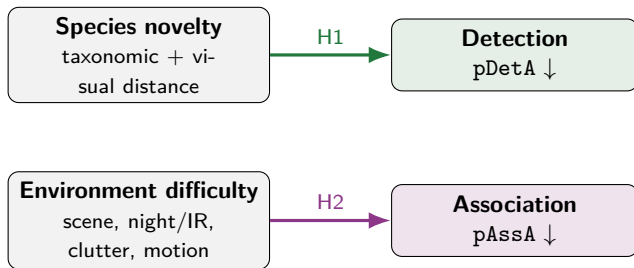
The metric decomposes — and that is the point



We analyse the **two components separately** (HOTA-style) — the hypothesis is they transfer for *different reasons*.

Hypotheses

H0 (null): the label-free distances carry *no* predictive signal — transfer isn't distance-driven. H1/H2 are the directional alternatives:



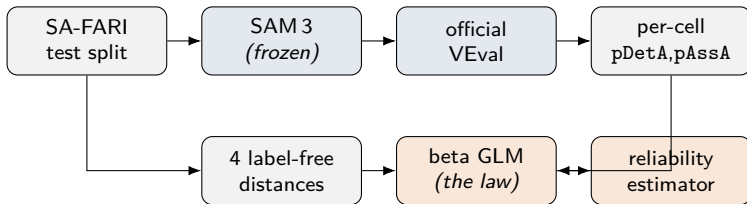
Failing to reject H0 is itself a result — it motivates a pivot to representational probing.

The data — SA-FARI: a natural experiment

- Largest open multi-animal wildlife tracking set (Meta $\times$ Conservation X Labs).
- **99 species** (full 7-level taxonomy) · **741 locations** / 4 continents · a decade of footage.
- **Train / test disjoint by species *and* location** $\Rightarrow$ every test video is a *controlled probe* of transfer along a measurable distance.
- Hard negatives (query returns nothing) are **kept** — they drive false-positive analysis.

Species $\times$ place $\times$ time = the axes we measure distance along.

Approach — freeze the model, fit a small law



SAM 3 **frozen**; the metric is the **official** one. Only the small **beta GLM** is fit — from label-free distances. **No target label enters a distance.**

The four label-free distances (+ a proxy)

- **Taxonomic** — tree steps to the nearest training species (lowest common ancestor).
- **Visual** — DINOv2 embedding of the *mask-cropped* animal vs. nearest training prototype (CLIP variant as a robustness check).
- **Environment** — scene embedding of the masked-out background vs. nearest training location; + a night/IR flag.
- **Temporal** — years to the nearest relevant training footage.
- **SAM-3 familiarity proxy** — separability in SAM 3's *own* feature space (hedges the unknown pretraining set).

All computed against the **frozen training split only** — the leakage firewall.

Novelty — stated honestly

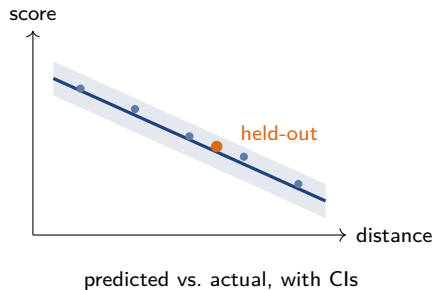
Not new: that OOD performance is predictable (for *classifiers*: Accuracy- & Agreement-on-the-Line); taxonomy in embeddings (BioCLIP); location shift (Terra Incognita).

Novel — the combination

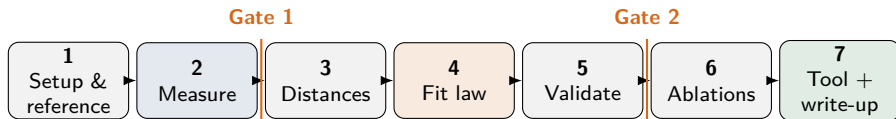
First predictable-transfer study for a **promptable video tracker**, **decomposed into two predictors** (detection vs association), on SA-FARI's species $\times$ place $\times$ time natural experiment — a deployment “**will it work here?**” estimator.

Validation & honesty

- Hold out **whole species & whole locations** (group-aware CV).
- Predict pDetA/pAssA from *distance alone*; report **bootstrap CIs** everywhere.
- **Success** = a *validated* predictor, not a higher score.
- **Gate**: no out-of-sample signal $\rightarrow$ invoke H_0 , pivot.



Implementation roadmap — seven steps, two gates



Gate 1 (after measurement): are the scores in SA-FARI's SAM 3 ballpark? **Gate 2** (after validation): real out-of-sample signal — else invoke H_0 and pivot.

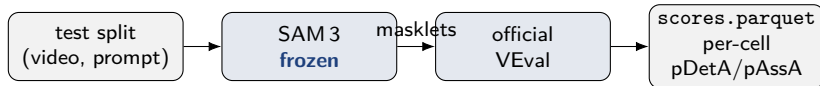
Step 1 — Setup & the leakage firewall (Phase 0)

- **Acquire SA-FARI** — gated annotations (Hugging Face) + public 6 fps frames (GCS); smoke-test SAM 3 + the official VEval on 1–2 clips.
- **Freeze the seen/unseen reference** — enumerate train species, locations, and taxonomy, then **assert** the test split is disjoint by species *and* location.

The leakage firewall

Every distance is later computed against the **frozen training set only** (data/reference/*) — no test information can enter a feature.

Step 2 — Measurement: frozen SAM 3 → VEval (Phase 1)



- Species-specific prompts (primary) + a generic “animal” prompt (robustness); hard negatives kept.
- Output = the **dependent variables**: per-cell pDetA/pAssA/pHOTA + support.

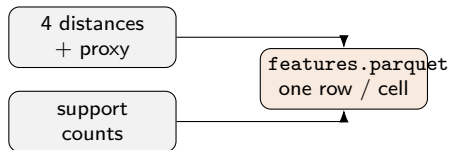
Gate 1: are the scores in SA-FARI's SAM 3 ballpark?

Step 3 — The four label-free distances (Phase 2)

Distance	computed from	column
Taxonomic	LCA tree steps to nearest train species	taxonomic_distance
Visual	DINOv2 mask-crop vs nearest prototype	visual_distance
Environment	scene embed of masked-out background	environment_distance
Temporal	years to nearest train footage	temporal_gap
Familiarity	separability in SAM 3's own features	familiarity_proxy

All **label-free** for the target and computed against the **frozen train set**.

Step 3 — Assemble the feature table (Phase 2)



Leakage unit test

Swapping the frozen seen set changes the distances *as expected* — no test label enters any column.

Step 4 — Fit the law & decompose (Phase 3)

- Model pDetA and pAssA **separately** with a **beta / logit-link GLM**, *weighted by support*, with a $\log(n_frames)$ covariate (so “rare” $\neq$ “far”) $\rightarrow$ outputs/models/*.
- **Variance partitioning** (dominance/Shapley + VIF) attributes each factor's contribution.

The decomposition test

Does detection load on **species novelty** and association on **environment**? — the headline contrast.

Step 5 — Out-of-sample validation (Phase 4)

- **Grouped CV** — hold out *whole* species and *whole* locations; predict held-out cells from **distance alone** → `cv_results.parquet` (MAE/RMSE, calibration).
- **Bootstrap CIs** everywhere + the **predictive-line** figure (predicted vs. actual).

Gate 2: real out-of-sample signal? — else invoke **H0** and pivot to representational probing.

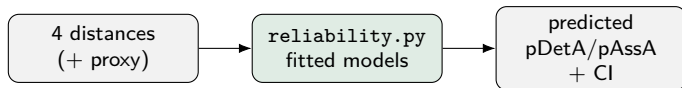
Success = a *validated* predictor, not a higher score.

Step 6 — Ablations & robustness (Phase 5)

- **Factor ablations** — drop each distance and report the R^2 / OOS-error change: what does each factor contribute?
- **Design & confound robustness** — DINOv2 vs. CLIP; mask-cropped vs. whole-frame; species vs. generic prompt; $\pm$ the support covariate.

The detection/association contrast **holds across variants** — or its limits are documented.

Step 7 — The reliability tool & write-up (Phase 6)



- The deployment “**will it work here?**” tool: a new (species, place) → a prediction + interval.
- The **dissertation** — every claim backed by a repo-generated figure/table.

Contribution

The deliverable: a **validated, label-free reliability estimator** for a promptable tracker
— and *why* detection and association transfer differently.

Thank you — questions welcome.